MediSense

AI CLINICAL DECISION SUPPORT

The AI assistant that diagnoses alongside the doctor.

A clinical intelligence platform that learns from real diagnostic histories to suggest differential diagnoses, recommend safe prescriptions, and prioritize the patients who need attention first.

Differential Diagnosis Engine

Drug Safety & Interaction Checking

Acuity-Based Triage

Hospital / EHR Integration

Product & Technical Documentation

Confidential — for investors & clinical partners

Version 1.0 · June 2026

First deployment: Hospital pilot, China

CONTENTS

What's inside this document

A complete product and technical reference for MediSense — written for clinical partners evaluating the product and investors evaluating the opportunity. It moves from the problem and product vision, through detailed features and user experience, into the AI architecture, data, integration, and safety model that make it work.

01 · EXECUTIVE SUMMARY

The product in one page

MediSense is an AI-powered clinical decision-support platform that sits beside the physician at the point of care. It learns from a curated knowledge base of real diagnostic episodes — patient symptoms, the doctor's diagnosis, the treatment and prescription that followed, and how well the patient recovered — and uses that knowledge to assist with the next patient.

When a new patient presents, MediSense reads their symptoms and history, compares them against thousands of similar past cases, and returns a ranked list of probable conditions with a confidence level and a transparent explanation of *why*. It then proposes evidence-aligned next steps: tests to confirm, and medications to consider — each one automatically screened against the patient's current drugs, allergies, and history so the doctor never has to chase a hidden interaction. Across a full ward or clinic, it continuously ranks patients by how urgent and critical their condition is, so the most fragile patient is never waiting at the back of the queue.

3,000+
SEED DIAGNOSTIC EPISODES IN THE INITIAL KNOWLEDGE BASE

4
CORE PILLARS: DIAGNOSE · PRESCRIBE · TRIAGE · INTEGRATE

<5s
TARGET TIME FROM SYMPTOM ENTRY TO RANKED SUGGESTIONS

100%
OF RX SUGGESTIONS SCREENED FOR INTERACTIONS & ALLERGIES

> **The core principle: assist, never replace.**
> MediSense never makes an autonomous medical decision. Every output is a **suggestion with its reasoning attached**, presented for the physician to accept, modify, or reject. The doctor's judgement is always the final authority — MediSense exists to make that judgement faster, safer, and better-informed.

Who it serves and why it matters

The first deployment is a hospital in China, where the system is trained and refined on Chinese clinical data and disease prevalence before expanding. For physicians, it reduces cognitive load and the risk of missed differentials. For hospitals, it standardizes quality, shortens decision time, and creates a learning asset that improves with every case. For patients, it means faster, safer, more consistent care.

02 · THE PROBLEM & OPPORTUNITY

Why clinicians need a second mind

Medicine is judgement under pressure. A physician sees dozens of patients a day, each with incomplete information, overlapping symptoms, and a medication history that may hide a dangerous interaction. Knowledge is vast and growing; attention and time are not.

Diagnostic uncertainty

Many conditions share symptoms. Rare or atypical presentations are easy to miss, and a single anchoring assumption early in the consult can steer the whole diagnosis off course.

Information overload

Guidelines, drug data, and prior records are scattered across systems. Synthesizing them for every patient, in minutes, is beyond what any individual can do reliably all day.

Medication risk

Drug–drug and drug–allergy interactions are a leading cause of preventable harm. Catching them requires a full, current view of the patient's regimen — which is rarely at hand.

Triage under load

In a busy ward, the sickest patient isn't always the loudest. Without a consistent way to rank acuity, urgent cases can wait while stable ones are seen.

The opportunity

Every one of these problems is, at its heart, a pattern-matching and synthesis problem — exactly what modern AI does well, provided it is grounded in real, outcome-labelled clinical data. Hospitals already generate that data with every consultation; today it is filed and forgotten. MediSense turns the institution's own accumulated experience into a living assistant that gets sharper with every patient it sees.

> **The asset hiding in plain sight**
>
> A hospital's archive of **symptoms** → **diagnosis** → **treatment** → **outcome** is the most valuable and underused dataset it owns. MediSense is the system that finally puts it to work for the next patient.

03 · PRODUCT VISION & PRINCIPLES

What MediSense is — and is not

MediSense is an **intelligent clinical co-pilot**: a system trained on prior medical knowledge that observes a new patient's symptoms, proposes the most probable diagnoses, recommends safe treatment and prescriptions, and helps the doctor decide who to see first.

What it IS

- A **decision-support** tool that augments the physician.
- A **transparent** reasoner that shows evidence behind every suggestion.
- A **safety net** that screens every prescription for interactions.
- A **learning system** that improves from real outcomes.
- A **connected** layer that reads the hospital's existing records.

What it is NOT

- **Not** an autonomous doctor — it never decides alone.
- **Not** a replacement for examination, labs, or imaging.
- **Not** a black box — no suggestion arrives without a reason.
- **Not** a static model — it is refined continuously.
- **Not** a data silo — patient data stays governed and secure.

Design principles

1 · Physician-in-command

The interface always frames AI output as a suggestion to be confirmed. The doctor accepts, edits, or overrides with one tap, and every override teaches the system.

2 · Explainable by default

Confidence scores and the specific symptoms, past cases, and rules driving each suggestion are always one tap away — never hidden.

3 · Safety over completeness

When data is missing or signals conflict, MediSense says so and asks, rather than guessing. It flags uncertainty loudly.

4 · Fast at the point of care

Sub-five-second suggestions, keyboard-light entry, and a layout designed for a clinician glancing between patient and screen.

04 · USERS & PERSONAS

Who we build for

MediSense serves several roles inside the hospital. The product is designed primarily around the attending physician, with dedicated surfaces for nurses, pharmacists, and administrators.

Dr. Li Wei — Attending Physician Primary

Goal: diagnose accurately and quickly, prescribe safely, move to the next patient.

Frustrations: fragmented records, fear of missing a rare condition, manual interaction checks.

MediSense gives: ranked differentials with evidence, auto-screened prescriptions, full history in one view.

Nurse Zhang — Triage & Ward Nurse Key

Goal: capture vitals and symptoms accurately, surface the patients who can't wait.

Frustrations: judging acuity under pressure, re-entering data.

MediSense gives: a live, auto-ranked patient queue and structured intake.

Pharmacist Chen — Clinical Pharmacist Key

Goal: verify that every prescription is safe and appropriate before dispensing.

Frustrations: incomplete medication lists, time-consuming manual checks.

MediSense gives: a pre-screened Rx with interaction flags and rationale to verify.

Director Wang — Medical Administrator Stakeholder

Goal: raise care quality and consistency, manage risk, demonstrate outcomes.

Frustrations: variable quality, limited visibility into diagnostic performance.

MediSense gives: analytics on accuracy, override rates, and outcome trends.

05 · CAPABILITY OVERVIEW

The four pillars

Everything MediSense does rests on four connected capabilities. Each is useful alone; together they form a continuous loop from a patient arriving to a safe, prioritized plan of care.

PILLAR 01	PILLAR 02	PILLAR 03	PILLAR 04
Diagnose	**Prescribe**	**Triage**	**Integrate**
Rank probable conditions from symptoms & history with confidence and evidence.	Recommend treatment & medication, auto-screened for interactions.	Score acuity & urgency to order the patient queue.	Read hospital & patient records; write back the agreed plan.

Pillar	What it does	Primary user	Emphasis
Diagnosis Engine	Compares new symptoms to the case knowledge base and returns ranked differentials with probability and rationale.	Physician	Core
Prescription & Safety	Suggests treatment and drugs; checks every option against the patient's drugs, allergies, age, renal/ hepatic status.	Physician, Pharmacist	Core
Triage	Continuously scores and ranks patients by criticality and urgency across the ward or clinic.	Nurse, Physician	Supporting
Integration	Connects to hospital systems (HIS/EHR) to pull history and write the confirmed diagnosis and Rx.	System / IT	Supporting

Why the loop matters

Diagnosis informs prescription; prescription depends on the integrated record; the record feeds triage; and the doctor's confirmed outcome flows back into the knowledge base — sharpening the next diagnosis. MediSense is a **closed learning loop**, not four separate tools.

06 · CORE CAPABILITY

Diagnosis & Differential Probability Engine

The heart of MediSense — turning a patient's symptoms into a ranked, explained set of probable diagnoses.

The physician describes or selects the patient's symptoms. The engine analyses them against the knowledge base of prior episodes and the patient's own record, then returns a **differential** — an ordered list of candidate conditions, each with a probability, a confidence band, and the evidence behind it.

How a diagnosis is produced

1 · Symptom capture & normalization

Free-text or structured symptom entry is mapped to a standard clinical vocabulary (e.g. SNOMED CT / ICD-aligned terms), with onset, duration, severity, and negatives ("no fever") captured explicitly.

2 · Patient-context enrichment

Age, sex, vitals, chronic conditions, current medications, allergies, and recent labs are pulled from the integrated record to condition the analysis.

3 · Similarity & probabilistic matching

A hybrid model retrieves the most similar past cases and combines them with a learned classifier and disease-prevalence priors to estimate the probability of each candidate condition.

4 · Ranking & calibration

Candidates are ranked and their confidence calibrated, so a stated "78%" reflects real-world frequency. "Cannot-miss" critical conditions are surfaced even at lower probability.

5 · Explanation

For each candidate the engine shows the supporting symptoms, contradicting signals, similar historical cases, and what additional test would most increase certainty.

What the doctor sees

Ranked differential	Each condition with a probability bar, confidence label (High / Moderate / Low), and a one-line "because…" summary.
Evidence drawer	Tap any condition to see matching symptoms, the count of similar past cases, typical outcomes, and contradicting findings.
Next-best-test	The single test or question that would most reduce uncertainty, so the doctor knows what to check next.
Red-flag banner	If symptoms match a time-critical condition (e.g. sepsis, MI, stroke), a prominent alert appears regardless of rank.

Anti-anchoring by design

The engine deliberately surfaces a **broad differential**, not a single answer, and always includes plausible "do-not-miss" alternatives. This counteracts the human tendency to lock onto the first plausible diagnosis.

Example output (illustrative)

Candidate condition	Probability	Confidence	Key supporting evidence
Community-acquired pneumonia	78%	High	Fever, productive cough, focal crackles, raised CRP; 142 similar cases
Acute bronchitis	41%	Moderate	Cough, low-grade fever; absence of focal signs lowers rank
Pulmonary embolism do-not-miss	9%	Watch	Pleuritic pain + tachycardia; surfaced despite low probability

Illustrative example for documentation purposes only. Probabilities and confidence are produced per-patient and per-deployment from the trained model and are always presented for physician review.

07 · CORE CAPABILITY

Prescription & Drug-Interaction Safety

From a confirmed diagnosis to a safe, personalized treatment and prescription — with every drug screened automatically.

Once the doctor confirms a diagnosis, MediSense proposes a treatment plan and candidate medications drawn from how similar cases were successfully treated. Critically, **every suggested drug is screened in real time** against the patient's full medication list, allergies, age, weight, and renal/hepatic function before it is ever shown.

The safety screen — what every suggestion passes through

Drug–drug interactions

Each candidate is checked against every medication the patient currently takes; contraindicated or risky combinations are blocked or flagged with severity.

Allergy & intolerance

Documented allergies and prior adverse reactions are matched against the drug and its class, including cross-reactivity.

Dose for the patient

Dosing is adjusted for age, weight, and renal/hepatic function; pediatric and geriatric limits are enforced.

Condition & context flags

Pregnancy, comorbidities, and duplicate-therapy checks (two drugs of the same class) raise contextual warnings.

Nothing unsafe reaches the doctor unflagged

A medication that fails a hard safety rule is never silently suggested. It is either withheld with an explanation or surfaced with a clear **severe interaction** banner and a safer alternative offered in its place.

Interaction severity model

Severity	Behaviour in the UI	Example
Contraindicated	Blocked; requires explicit override with reason; alternative offered.	Drug + drug with life-threatening combined effect
Major	Prominent warning; doctor must acknowledge before prescribing.	Significant additive risk needing monitoring
Moderate	Inline flag with guidance (e.g. adjust dose, monitor labs).	Manageable interaction with precautions
Minor / info	Subtle note, no interruption.	Low-significance or theoretical interaction

The prescription flow

STEP 1
Confirm Dx
Doctor accepts a diagnosis from the differential.

›

STEP 2
Suggest plan
Treatment & drug options from similar successful cases.

›

STEP 3
Auto-screen
Each option checked vs. record; risky ones flagged.

›

STEP 4
Doctor decides
Accept, swap, or adjust; override is logged.

›

STEP 5
Sign & send
Rx written to the record & pharmacy.

08 · SUPPORTING CAPABILITY

Triage & Patient Prioritization

Across a busy ward or clinic, MediSense maintains a live, ordered queue of patients ranked by an **acuity score** that blends how critical and how urgent each patient's condition is — so the doctor always knows who needs attention first.

What feeds the acuity score

- Vital signs and early-warning thresholds (e.g. respiratory rate, SpO_2, blood pressure, heart rate, temperature).
- Severity of the top candidate diagnoses and presence of any "do-not-miss" red flags.
- Trend — is the patient deteriorating or stable since the last reading?
- Time waiting, age, and known high-risk comorbidities.

Priority bands

CRITICAL Immediate — possible life threat; pushed to top, alert raised.

URGENT See soon — significant risk or deteriorating trend.

ROUTINE Stable — standard queue order.

Bands are visually unmistakable and re-rank automatically as new vitals or results arrive.

Escalation, not just ordering

When a patient crosses into **Critical**, MediSense does more than re-sort the list — it raises an active alert to the responsible clinician so a deteriorating patient is never missed because someone wasn't looking at the screen.

The triage board (illustrative)

#	Patient	Chief complaint	Key vitals	Top suggestion	Priority
1	Patient A, 67M	Chest pain, sweating	HR 118, BP 92/60	ACS — do-not-miss	CRITICAL
2	Patient B, 24F	Severe abdominal pain	Temp 39.1°C	Appendicitis	URGENT
3	Patient C, 41M	Cough, mild fever	SpO_2 97%	Bronchitis	ROUTINE

Illustrative example. Scores and bands are computed per-patient from live data and clinical rules and are always advisory to the care team.

09 · SUPPORTING CAPABILITY

Hospital & EHR Integration

MediSense is only as good as the data it sees. It connects to the hospital's existing systems to pull a complete, current patient picture — and to write the confirmed diagnosis and prescription back where the rest of the care team can act on it.

What it reads and writes

Reads from the hospital record

- Demographics, allergies, problem list, chronic conditions.
- Current and past medications (the basis of interaction checks).
- Recent vitals, lab results, and imaging reports.
- Encounter and visit history.

Writes back (after doctor sign-off)

- The confirmed diagnosis and coded problem.
- The signed prescription, routed to pharmacy.
- Ordered tests and follow-up plan.
- An audit trail of what was suggested vs. chosen.

How it connects

Standards-based interfaces

Primary integration via **HL7 FHIR** resources (Patient, Condition, MedicationRequest, Observation, AllergyIntolerance) with HL7 v2 messaging where the hospital's HIS requires it.

Adapter layer for legacy HIS

A configurable adapter maps each hospital's local data formats and code systems to the MediSense canonical model, so onboarding a new site is configuration, not re-engineering.

China-first localization

The first deployment integrates with the partner hospital's HIS and is tuned to local terminology, drug formularies, and coding conventions; the architecture keeps these as swappable configuration for future sites.

Security at the boundary

All exchange is encrypted, authenticated, scoped to the minimum data needed, and fully logged. Data residency and governance follow the hospital's and jurisdiction's requirements.

> **Designed to fit in, not rip and replace**
>
> MediSense layers on top of the systems a hospital already runs. The physician keeps their existing workflow; MediSense adds intelligence inside it rather than asking the institution to migrate.

10 · END-TO-END USER FLOW

The doctor's journey, start to finish

A single consultation, walked through end to end — showing where MediSense assists and where the physician stays firmly in command.

① Patient arrives & is registered

Nurse captures chief complaint and vitals at intake. MediSense immediately places the patient in the triage queue with a provisional priority band.

② Doctor opens the patient

A unified patient view loads: integrated history, current medications, allergies, and recent results — no hunting across systems.

③ Symptoms entered

The doctor enters or refines symptoms. The engine returns a ranked differential within seconds, with red flags surfaced first.

④ Doctor reviews the evidence

They open the evidence drawer on the leading candidates, check supporting and contradicting signals, and see the suggested next-best test.

⑤ Tests ordered (optional)

If confirmation is needed, tests are ordered through the integration; results flow back and update the differential automatically.

⑥ Diagnosis confirmed

The doctor accepts a diagnosis (or records their own). The choice — including any override of the AI's top suggestion — is logged for learning.

⑦ Treatment & prescription

MediSense proposes a plan and medications, each pre-screened for interactions, allergies, and dosing. The doctor adjusts and signs.

⑧ Plan written back

The signed diagnosis, prescription, and follow-up are written to the hospital record and routed to pharmacy and the care team.

⑨ Outcome & feedback

Recovery feedback and outcome are captured later and fed back into the knowledge base, improving suggestions for the next similar patient.

The human checkpoint is non-negotiable

Steps ⑥ and ⑦ — confirming the diagnosis and signing the prescription — **always require explicit physician action**. MediSense can prepare everything, but the doctor commits it.

11 · SCREEN-BY-SCREEN UX

Key interfaces

Wireframe-level layouts of the four screens a physician lives in. The design language favours clarity, glanceability, and keeping AI output visibly framed as a suggestion.

A · Triage board — the daily home screen

MediSense · Triage Board — Internal Medicine | **Dr. Li · 12 patients**

FILTERS

DEPARTMENTS

CRITICAL · 2 | **URGENT · 4** | **ROUTINE · 6**

Auto-sorted by acuity. Critical rows pulse and raise an alert.

B · Patient & diagnosis view — the consultation screen

Patient A · 67M · MRN 0098-2231 | **Allergies: Penicillin**

HISTORY & MEDS

CURRENT DRUGS

⚠ RED FLAG: possible ACS — assess now

DIFFERENTIAL

Each row: condition · probability bar · confidence · "because…" · evidence ›

C · Prescription & safety

Prescribe — CAP | **screened**

⚠ Penicillin allergy → alt. suggested

Every drug pre-checked; severe flags block until resolved.

D · Evidence drawer

Why pneumonia? 78% | **142 cases**

Supports: fever · cough · crackles · ↑CRP

Against: no pleuritic pain

Next-best test: chest X-ray (+confidence)

One consistent interaction model

> Across every screen, AI output uses the same visual grammar: a coloured confidence cue, a plain-language reason, and an always-available "show evidence." The doctor learns to trust it because it always shows its work.

How the model thinks

MediSense uses a **hybrid architecture** — combining case-based retrieval, machine-learned classification, and an explicit clinical rules layer. No single technique is trusted alone; each covers the others' blind spots, and the rules layer guarantees safety.

The three layers

1 · Retrieval (case-based)

Each past episode is embedded into a vector representation of its symptoms and context. A new patient retrieves the most similar historical cases — the engine reasons from real precedent, and can always point to "patients like this one."

2 · Learned classifier

A supervised model trained on the labelled episodes estimates the probability of each condition from the symptom/context vector, capturing patterns beyond any single neighbour and calibrating the final probabilities.

3 · Clinical rules & safety

An explicit, expert-curated rules layer enforces red-flag detection, interaction/allergy blocks, and dosing limits. Rules can hard-override the statistical layers — safety is never left to probability alone.

The reasoning pipeline

```
# Conceptual flow — symptoms in, explained suggestions out
patient = enrich(symptoms, ehr_record)        # context: age, meds, labs, allergies
neighbors = retrieve(patient, knowledge_base)   # most similar past episodes
probs = classify(patient, neighbors)            # calibrated disease probabilities
probs = apply_rules(probs, patient)            # red flags + do-not-miss surfacing
dx = rank_and_explain(probs, neighbors)       # ordered differential + evidence
rx = recommend_treatment(dx, knowledge_base)
rx = safety_screen(rx, patient.meds, patient.allergies)  # interactions, dose, allergy
return dx, rx   # presented to physician for confirmation
```

Why hybrid, not a single black box

Property	How the architecture delivers it
Explainability	Retrieval gives "cases like this"; rules give named reasons. Every suggestion traces to evidence.
Safety	The rules layer can override statistics — hard constraints (allergy, contraindication) always win.
Cold-start tolerance	For rare conditions with few examples, retrieval + rules still function where a pure classifier would be unreliable.
Calibration	Probabilities are calibrated against observed frequencies, so confidence is honest.
Improvability	New labelled episodes update retrieval and classifier without re-engineering the rules.

Handling uncertainty honestly

> When evidence is thin or signals conflict, the engine widens its differential, lowers stated confidence, and explicitly asks for the missing information rather than projecting false certainty.

13 · KNOWLEDGE BASE & CONTINUOUS LEARNING

The feedback loop that makes it smarter

MediSense begins with a seed knowledge base of **3,000+ diagnostic episodes** and grows with every consultation. Each episode is a complete, outcome-labelled record — the unit the whole system learns from.

Anatomy of a knowledge-base episode

① Symptoms & presentation

Structured symptoms with onset, severity, and negatives, plus patient context at the time of the visit.

② Doctor's diagnosis

The confirmed condition(s), coded to a standard vocabulary — the ground-truth label.

③ Treatment & prescription

The solution the doctor chose: procedures, medications, doses, and follow-up.

④ Recovery feedback

How well the patient improved — the outcome signal that tells the model which solutions actually worked.

> **Outcome is the secret ingredient**
>
> Most systems learn "symptoms → diagnosis." MediSense also learns "treatment → **recovery**." Weighting suggestions by what genuinely led to good outcomes is what lets it recommend not just a plausible plan, but a plan that has *worked* for patients like this one.

The continuous-learning loop

CAPTURE		CURATE		LEARN		EVALUATE		SERVE
New episode	›	**Validate**	›	**Update model**	›	**Measure**	›	**Improve care**
Each confirmed case is structured and stored.		Quality checks, de-identify, code, review.		Retrieval index & classifier refreshed.		Accuracy & calibration re-checked before release.		Sharper suggestions for the next patient.

Quality & governance of the data

- **Human-validated labels:** diagnoses and outcomes come from physicians, not inferred guesses.
- **Bias monitoring:** performance is tracked across age, sex, and condition groups to catch skew before it harms care.
- **Versioned models:** every model release is versioned and evaluated against a held-out set; regressions block release.
- **De-identification:** data used for learning is governed and minimized per the security model in §15.

The foundation underneath

A layered architecture keeps the clinical interface fast, the AI services independently scalable, and the integration with hospital systems cleanly separated behind an adapter.

Logical architecture

- **Presentation layer**
  Clinician web/desktop app — triage board, patient view, diagnosis & prescription surfaces. Designed for speed and glanceability.
- **Application & API layer**
  Orchestrates a consultation: auth, session, request routing to AI services, write-back, and audit logging.
- **AI services**
  Independent services for retrieval, classification, rules/safety screening, and triage scoring — each separately deployable and scalable.
- **Data & knowledge layer**
  The episode knowledge base, vector index, drug-interaction reference, and the canonical patient model.
- **Integration adapter**
  FHIR/HL7 connectors mapping each hospital's HIS to the canonical model — the single place site-specific differences live.

Core data entities

Entity	Key attributes
Patient	ID, demographics, allergies, problem list, chronic conditions (from EHR via adapter)
Encounter	Visit context, presenting complaint, vitals, attending clinician, timestamp
SymptomSet	Coded symptoms with onset, severity, duration, and explicit negatives
DiagnosisEpisode	Symptoms · doctor's diagnosis · treatment/Rx · recovery outcome — the KB unit
Medication	Drug, class, dose, route; current & historical, for interaction screening
Suggestion	Ranked differential / Rx options with probability, confidence, evidence refs
Decision	What the doctor confirmed vs. what was suggested, with override reason & audit
AcuityScore	Computed priority band, contributing factors, timestamped trend

> **The Decision entity is doubly valuable**
> By recording suggestion *and* the doctor's final choice, MediSense builds both an **audit trail** for safety and governance and a **training signal** — every physician override is a lesson the system learns from.

15 · SECURITY, PRIVACY & COMPLIANCE

Trust & safety

A clinical system handling patient data must earn trust before anything else. Security, privacy, and clinical safety are built into the architecture — not added afterward.

Data protection

- Encryption in transit and at rest across all services.
- Role-based access control; clinicians see only what their role permits.
- Data minimization — only the fields needed for a task are exchanged.
- De-identification for data used in model learning.
- Configurable data residency to meet local jurisdiction rules.

Accountability

- Full audit trail of every suggestion, view, and decision.
- Immutable logging of overrides and prescription sign-offs.
- Versioned models — every output is traceable to a model version.
- Clear attribution: actions are tied to an authenticated clinician.

Clinical safety guardrails

Human-in-the-loop, always

No diagnosis is recorded and no prescription is issued without explicit physician confirmation. The system cannot act autonomously on a patient.

Fail-safe defaults

On low confidence, missing data, or conflicting signals, MediSense defers to the clinician and asks — it never fills gaps with confident guesses.

Hard safety rules win

Allergy and contraindication blocks override any statistical suggestion; unsafe options are withheld or loudly flagged.

Continuous monitoring

Accuracy, calibration, override rates, and subgroup performance are monitored so drift or bias is caught early.

Regulatory posture

MediSense is positioned as **clinical decision support**, keeping the licensed physician as the decision-maker. Each deployment market — beginning with China — has its own medical-device and data-protection requirements; the product is built so that validation, documentation, and data-governance evidence can be produced for the relevant regulator of each market it enters.

16 · DESIGN SYSTEM

The visual language

A calm, clinical, high-contrast visual system built for fast reading at the point of care, where colour carries meaning — never decoration.

Colour & meaning

Teal · Primary
Brand, navigation, AI confidence-positive.

Indigo · Action
Prescription & interactive actions.

Emerald · Safe
Cleared checks, routine priority, good outcomes.

Amber · Caution
Urgent priority, moderate warnings.

Red · Critical
Red flags, contraindications, critical acuity.

Ink · Text
Primary text and high-contrast labels.

Principles in the interface

Glanceability
Priority and confidence are readable in under a second from across a room. Status is shown by colour *and* label, never colour alone — accessible to colour-vision deficiency.

Suggestion framing
AI output always carries a confidence cue, a plain-language reason, and an "accept / edit / reject" control — visually distinct from confirmed clinician input.

Progressive disclosure
The summary is always visible; the evidence and detail are one tap away. The doctor controls depth.

Low-friction entry
Keyboard-light, autocomplete-driven symptom and drug entry, optimized for speed during a live consult.

Bilingual interface (Chinese / English) is a first-class requirement for the China deployment; all components are built for localization from day one.

17 · ROADMAP & ROLLOUT

China-first, then scale

A staged rollout that proves safety and value in one hospital before broadening — de-risking each step with evidence before the next.

Phase 1 · Foundation & pilot China hospital

Train on the 3,000+ Chinese episode seed set; integrate with the partner hospital's HIS; deploy the diagnosis engine and drug-safety screen with a focused group of physicians. Goal: prove accuracy, safety, and clinician trust.

Phase 2 · Full clinical workflow

Add the triage board, outcome-feedback capture, and the continuous-learning loop. Expand to more departments in the pilot hospital. Goal: demonstrate measurable workflow and quality gains.

Phase 3 · Multi-site & specialty depth

Onboard additional hospitals via the integration adapter; deepen specialty coverage; mature analytics for administrators. Goal: prove the model generalizes across sites.

Phase 4 · Scale & new markets

Expand beyond the initial region with localization of formularies, language, and regulatory evidence per market. Goal: a repeatable deployment playbook.

Earn trust before scale

Each phase has explicit accuracy, safety, and adoption gates. MediSense expands only when the evidence from the prior phase justifies it — protecting patients and the product's credibility alike.

18 · SUCCESS METRICS & RISKS

How we measure & de-risk

Success metrics

Metric	What it tells us
Top-1 / Top-3 diagnostic accuracy	How often the confirmed diagnosis appears as the engine's leading / top-three suggestion.
Calibration error	Whether stated confidence matches real-world frequency — honesty of the probabilities.
Interaction catches	Count of risky prescriptions flagged before they reached the patient.
Time-to-decision	Reduction in time from patient open to confirmed plan.
Physician acceptance rate	How often suggestions are accepted — a proxy for usefulness and trust.
Outcome improvement	Recovery-feedback trends for cases where MediSense assisted.

Key risks & mitigations

Risk	Mitigation
Over-reliance / automation bias	Suggestion framing, mandatory confirmation, visible uncertainty, and override logging keep the doctor engaged.
Data bias or skew	Subgroup performance monitoring; diverse, validated training data; bias gates before model release.
Incorrect suggestion reaching a decision	Hard safety rules, do-not-miss surfacing, calibrated confidence, and human-in-the-loop at every commit.
Integration variability across hospitals	Adapter layer isolates site differences; standards-based FHIR/HL7 reduce bespoke work.
Privacy & regulatory exposure	Encryption, minimization, residency controls, audit trails, and a decision-support regulatory posture.
Clinician adoption	Fits existing workflow, fast UX, transparent reasoning, and phased rollout that earns trust early.

The through-line

Every risk is met by the same two commitments that define the product: **keep the physician in command**, and **always show the reasoning**. Those are not just principles — they are the safety mechanism.

19 · GLOSSARY

Key terms

Differential diagnosis — an ordered list of candidate conditions that could explain a patient's symptoms, rather than a single answer.

Acuity score — MediSense's computed measure of how critical and urgent a patient's condition is, used to order the triage queue.

Knowledge base — the collection of outcome-labelled diagnostic episodes (symptoms → diagnosis → treatment → recovery) the system learns from.

Episode — one complete past case in the knowledge base; the unit of learning.

Calibration — alignment between stated confidence and real-world frequency, so a "70%" means roughly 70% in practice.

Retrieval (case-based) — finding the most similar historical cases to reason from precedent and explain suggestions.

Clinical decision support (CDS) — software that assists a clinician's judgement without replacing it; the regulatory posture of MediSense.

Do-not-miss — a time-critical condition surfaced even at low probability because the cost of missing it is severe.

FHIR / HL7 — healthcare interoperability standards used to read from and write to hospital records.

HIS / EHR — Hospital Information System / Electronic Health Record; the systems MediSense integrates with.

Human-in-the-loop — the requirement that a clinician confirms every diagnosis and prescription before it takes effect.

Override — when a doctor chooses differently from the AI suggestion; logged for audit and learning.

MediSense

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